

Recitation 03
Linear Quadratic Regulator

Exercise 1

Re-engineer the parameter $q \in \mathbb{R}$ of the stage cost function

$$\ell(x, u; q) = qx^2 + u^2 \quad (1)$$

such that the solution of the infinite horizon optimal control problem

$$\min_{x_i, u_i} \sum_{i=0}^{\infty} \ell(x_i, u_i) \quad (2a)$$

$$\text{s.t. } x_{i+1} = x_i + u_i, \quad i \in \mathbb{N}^+ \quad (2b)$$

$$x_0 = x(k) \quad (2c)$$

is given by the control law $u(k) = -\frac{1}{3}x(k)$.

Exercise 2

Use linear quadratic regulator (LQR) design in order to approximately minimize the infinite horizon cost

$$J(x_0) = \sum_{k=0}^{\infty} x(k)^2 + \frac{3}{4}u(k)^2 \quad (3)$$

with respect to the nonlinear system

$$x(k+1) = x(k) + x(k)^2 + u(k), \quad x(0) = x_0, \quad (4)$$

and to investigate the resulting closed-loop stability properties.

1. Linearize (4) around the equilibrium point $x_s = 0$ and $u_s = 0$.
2. Design an LQR controller $u(k) = F_{\infty}x(k)$ based on the linearized system that approximately minimizes the infinite horizon cost (3).
3. Which stability properties of the origin $x_s = 0$ for the resulting linear closed-loop system can be directly concluded?
4. Which stability properties of the origin $x_s = 0$ for the resulting nonlinear closed-loop system can be concluded?